

Pneumonia Detection from Chest X-ray Images using Convolutional Neural Network

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Abstract—The fact that Pneumonia ranks among the world's most common causes of mortality, precise diagnosis methods are absolutely necessary. Although chest X-rays are a quick way to identify Pneumonia, they can be challenging to interpret due to the similarities between Pneumonia and other lung conditions. This research presents a computer-aided approach to Pneumonia diagnosis that uses chest x-rays to enhance diagnostic decision-making. Moreover, situations such as the coronavirus pandemic, where widespread lockdowns are implemented and human contact poses significant risks, highlight the importance of computer-aided diagnosis like this. Consequently, a technique for the quick and automatic diagnosis of Pneumonia is presented in this work. Based on chest X-ray pictures, a deep learning-based architecture called “ESPD” is suggested for the automatic diagnosis of Pneumonia. A benchmark dataset comprising 5,856 chest X-ray pictures was utilized for the suggested deep-learning network's testing, evaluation, and training. The total accuracy of the suggested model was found to be 98.24%, consisting of 0.98 F1-Score, 0.98 precision, 0.98 recall, and 0.97 specificity. In comparison to previous methods in the literature, the suggested method was found to be quicker and less computationally expensive, and it also produced accuracy that showed promise.

Keywords—Pneumonia, Chest X-ray, Data Augmentation, Convolutional Neural Network.

I. INTRODUCTION

Chronic infections of the lungs like Pneumonia affect an excessive number of young children around the globe every year. Pneumonia was the primary reason for among children under the tender age of five between 1990 and 2019 [1]. In areas with poor healthcare services, the most common cause of death for children and young adults is Pneumonia. The lack of a reliable healthcare system in many developing and middle-income countries compounds the problem. The elderly and those with prior health risks are particularly at risk from this condition. Rapid breathing, discomfort in the chest area, prolonged coughing, initial cold-like symptoms and difficulties maintaining normal breathing patterns are all signs of Pneumonia, as defined by the World Health Organization (WHO) [2].

Sustainability integrates the utilization of Convolutional Neural Network (CNN) for identifying cases of Pneumonia, combining both environmentally conscious procedures with medical evaluation [3]. The utilization of CNNs for imaging purposes boosts Pneumonia determination, resulting in rapid treatment. Industry 5.0 ensures environmentally friendly operations and environmentally compatible hardware options. This collaboration serves equally leading-edge medical services and sustainable practices research and development initiatives [3].

The laboratory preferred method for Pneumonia is radiography (chest X-rays). Notwithstanding this, psychiatrists continue analyzing images individually, in spite of being constrained due to variables like exhaustion and their lack of experience. It necessitates quite a bit of effort and financial resources to turn into an appropriately trained radiology [4]. The rural parts of a nation with low revenues additionally experience a lack of radiologists. With the resemblance to other illnesses, evaluating Pneumonia via an X-ray of the chest may prove challenging even to highly qualified radiologists. If a doctor gives the inaccurate diagnosis, for instance, the individual in question might miss the medical treatment they require and can die as the consequence. Thus, Pneumonia could be problematic to identify properly [5].

For all of these factors, developing of Computer Aided Diagnosis (CAD) to assist in determining the presence of Pneumonia is of the greatest possible significance [6]. In recent years, CAD methods based on Deep Learning (DL) are gaining acceptance for use in the field of medicine. CNN, a type of DL-based CAD system inspired by the visual cortex of people, are able to learn from a wide range of information, particularly data that has been appraised by psychiatrists [7]. This paper introduces a deep learning based architecture called ESPD (Early Stage Pneumonia Detection) specifically designed for the automatic diagnosis of Pneumonia from chest X-ray images. Our proposed model achieves a high accuracy of 98.24% on a benchmark dataset consisting of 5,856 chest X-ray images.

II. LITERATURE REVIEW

To identify Pneumonia, researchers have employed a variety of strategies and techniques. Some of the existing research works are described in this section.

The authors of [8] Fusion of convolutional neural networks based on Dempster-Shafer theory for automatic Pneumonia detection from chest x-ray images, used five pre-trained CNNs and the Dempster-Shafer principle which recommends an original approach for accurately recognizing Pneumonia from chest X-ray images. The outcomes of the fusion-based DL algorithm are discussed accompanying a detailed analysis of the accuracy of every part of the CNN classifier. According to other state-of-the-art methods, the suggested strategy has greater detection performance, as indicated by its high precision (97.5%), recall (98.0%), F1-score (97.8%) and accuracy (97.3%). The recourse of only five pre-trained CNNs is an eventual drawback of the fusion-based DL approach. Additional CNNs or other fusion-based DL techniques could further boost the performance of the suggested method.

In [9], the authors integrated information from the Josep-NIH and the GWCMC_x. There were a total of 5856 X-ray images found in the GWCMC_x dataset from a hospital, while the COVID-19 and NIH databases were used in the Josep-NIH dataset. These sets of data were utilized for model creation and evaluation. The most reliable and responsive model for GWCMC_x achieved accuracy, sensitivity values of 83.3%, 89.3% for radiation, 89.9%, 93.6% for spatial dimension and 91.31%, 90.5% for superpixels-based histon respectively. In regard to compassion, the best Josep-NIH models were 99.0% accurate and 100% sensitive to fractals that 99.0% accurate and 98.6% sensitive to histons based on superpixels and 95.3% accurate and 99.2% sensitive to radiomics. Model clarity was overlooked in the research. Additional research is required to establish accurate models which are straightforward for doctors to use.

A overview of prior work using DL to recognize lung infections from chest X-rays is presented in [10]. CheXNet, a 121-layer CNN model designed by Rajpurkar et al. using 100,000 X-ray images from 14 various illnesses, is discussed in this research. The CNN method improves the standard radiological Pneumonia detection. The suggested method shows remarkable performance, with an accuracy of 93.91% and a F1-score of 93.88% in testing. The research might have been more viable with the latest techniques when it included a contrast using group methods for learning. Assembling may boost performance by assigning different weights to models based on their precision.

To identify Pneumonia in X-ray pictures of the chest, the Multi-Branch Fusion Auxiliary Learning (MBFAL) method is introduced in [11]. At an accuracy of 95.61%, MBFAL outperforms baseline models, particularly for fast Pneumonia identification. The approach provides various but equivalent accuracy levels throughout an array of categories, including “Normal” (98.70%), “COVID-19” (99.10%), “Other Viral Pneumonia” (96.60%) and “Bacterial Pneumonia” (96.80%).

With the objective for increasing the procedure’s quality as well as accuracy in practical situations, this research took consideration of the impacts variables like age, gender and pathologies on its efficacy.

The use of deep transfer learning for Pneumonia identification in chest X-ray pictures is investigated in [12]. This tests various methods for transfer learning using a data set of 5856 annotation images, finding that Xception yields the best outcomes. Researchers examined into a variety of devices, such as DenseNet121, DenseNet169, ResNet50, ResNet101, MobileNetV2, VGG16, Xception and InceptionV3. At an acceptance rate of 96.16% and an evaluation reliability of 95.73%, Xception easily overtook the other contenders. Research shows that transfer learning models are very useful for classification tasks like Pneumonia diagnosis. It is advised that further study be conducted using transfer learning models for medical image classification employing hybrid deep neural network frameworks.

In order to identify Pneumonia from chest X-ray images, the authors of [13] introduces a novel approach to transparent Artificial Intelligence using an architecture that is hybrid utilizing the transformer’s ensembles technique. This framework beats each of the ensemble situations by 2.05% (for binary class) and 1.3% (for multiclass), respectively. This study combines appropriate research and analyzes the hybrid DL framework to facilitate the binary and multiclass identification techniques. There were two different chest X-ray datasets utilized for training and validation, namely Mendeley and chest X-ray. Normal, Bacterial Pneumonia and Viral Pneumonia were the classes addressed in the chest X-ray dataset, while Normal and Pneumonia were the classes covered in the Mendeley dataset. A variety of medical images were utilized to demonstrate the effectiveness of the technique via automated parameters adaptation and analysis of statistics.

III. RESEARCH METHODOLOGY

In this section, we have explained the techniques and processes of our proposed method. The flowchart of our suggested technique is depicted in Fig. 1. There are some steps in our research including gathering data, preprocessing data, training information, standardizing data, supplementing data, developing models, assembling models and testing models which are described in the following subsections.

A. Data Collection

We have collected our dataset from Kaggle [14]. This dataset includes images of “Normal” and “Pneumonia” chest X-ray. The dataset is illustrated in Fig. 2.

B. Data Preprocessing

There are many steps involved with the Machine Learning (ML) assignment including classification of images. Matplotlib, Seaborn, Keras and OpenCV are few of the libraries that have been incorporated in this research. The *ImageDataGenerator* function is used for data separation and enhancing images whereas *BatchNormalization* is used

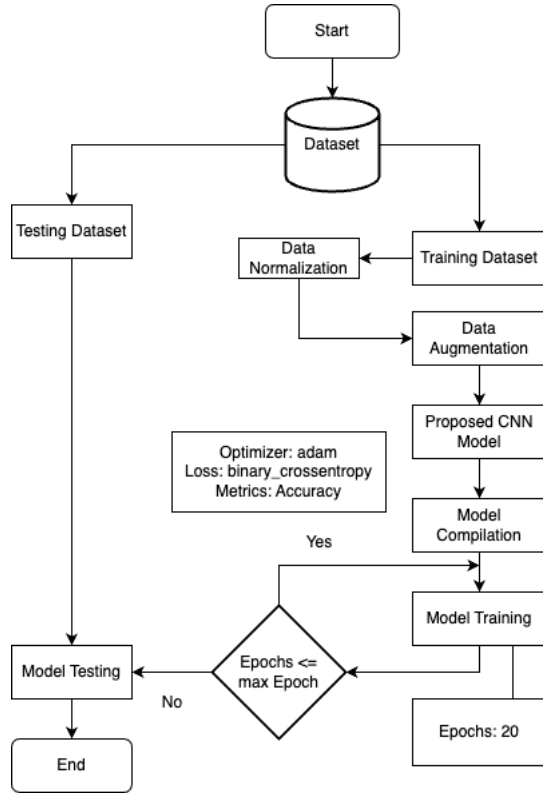


Fig. 1. Flowchart of the Proposed Methodology

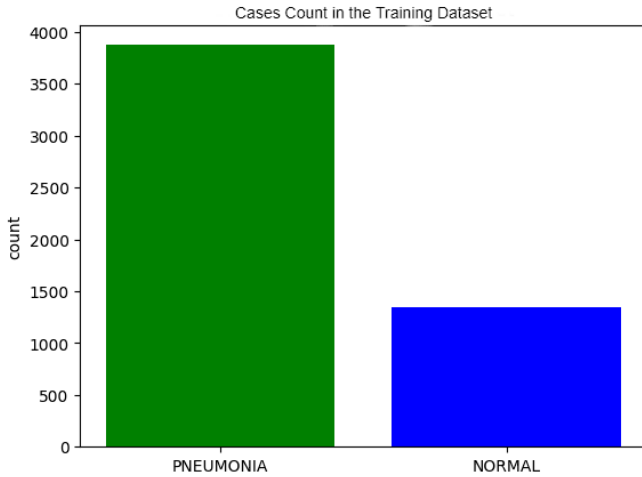


Fig. 2. Numbers of Data for Each Class for the Dataset [14]

for data normalization. Both functions are vital preparatory processes. Thihs's CNN model makes use of Conv2D, Max-Pool2D, Dense and Flatten layers in its design. In addition, the learning rate is adaptively adjusted using the callback function *ReduceLROnPlateau*. It then uses metrics like the classification report and confusion matrix to assess how well the model will perform.

1) *Data Training & Validation*: In ML, training the dataset is crucial since it includes teaching the system how to complete

a task or make reliable predictions. Data validation is required to check the accuracy of the model's predictions in light of new information. In this test, we have used photos that are reduced in size to 256×256 pixels. In Table I, the number of images that are used to test, train and validate for every class in our proposed model is given.

TABLE I
NUMBER OF IMAGES USED FOR THE PROPOSED MODEL

Class Name	Number of Images	Normal	Pneumonia
Test	630	234	396
Train	5,216	1,341	3,875
Validation	16	8	8
Total	5,862	1,583	4,279

2) *Data Normalization*: We have used color images with three separate channels and pixel values between 0 to 255 for this research. We have used grayscale normalization to compensate for lighting changes. To accomplish this normalization in the Keras framework, we have used a special equation, designated in Equation 1.

$$\text{normalization} = \text{layers.Rescaling} \left(\frac{1}{255} \right) \quad (1)$$

3) *Dataset Configuration*: We have fetched the data from Kaggle [14].

4) *Data Augmentation*: We have intentionally increased the size of our dataset to prevent the overfitting issue. We have employed flips (horizontal and vertical), height and width adjustments, rotation and zoom. We have added more training examples and developed a strong model by using these in our training data.

TABLE II
DATA AUGMENTATION SETTINGS

Method	Settings
Rescale	1/255
Rotation Range	30
Zoom Range	0.2
Width Shift Range	0.1
Height Shift Range	0.1
Horizontal Flip	True
Vertical Flip	False

C. Model Building

We have constructed a Sequential CNN architecture for this study, in which each layer receives a single input and generates a single output. These layers, which were carefully constructed, make up the entire network. We have employed a model with 1672449 total parameters in this configuration. An illustration of the CNN architecture utilized in this study is given in Fig. 3.

D. Model Compilation

Keras's building of models is required for the use of an Optimizer, a Loss Function and some Metrics. Compilation is the last stage of model construction. The model is compiled

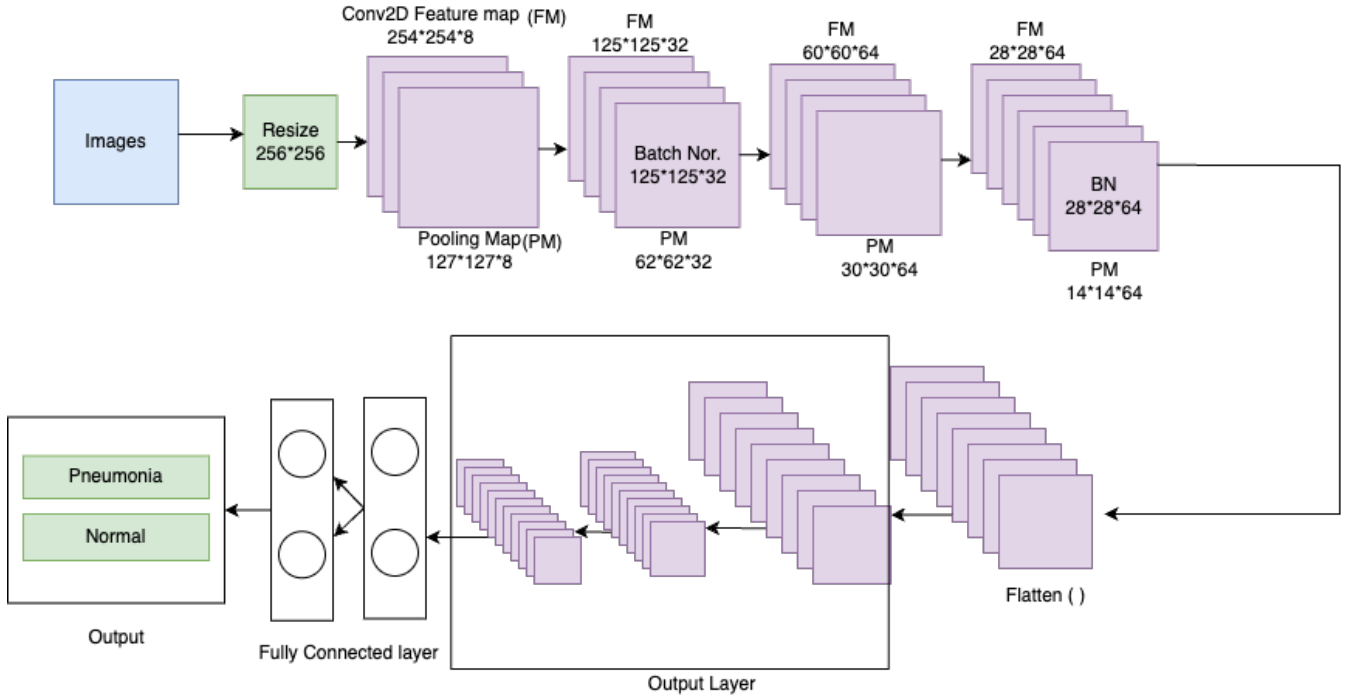


Fig. 3. Proposed CNN Architecture

without the use of weights. Changes of the model compilation or the experimentation with different compiling approaches will not affect the pre-trained weights. Before a model can be trained, its compilation must be complete. When the compilation is done, the training process can begin.

1) *Optimizer*: Optimizers are used to change parameters such as learning rates and weights in order to maximize the performance of neural networks. Adaptive Moment Estimation, or Adam for short, is a well-liked optimization technique that is frequently applied to machine learning model training. Adam integrates momentum and RMSprop, two more optimization methods.

- **Adaptive Learning Rates**: Adam's adaptive learning rate method is one of its primary characteristics. It uses the historical gradients to modify the learning rates of each parameter separately. Adam can effectively manage non-stationary targets and sparse gradients thanks to its adaptability.
- **Momentum**: Adam incorporates the concept of momentum to accelerate the optimization process. The algorithm maintains an exponentially decaying average of past gradients, which helps in smoothing out the optimization process and handling noisy gradients.
- **Bias Correction**: Adam addresses the issue of bias in the estimation of the first and second moments by incorporating bias correction. This correction ensures that the moving averages of the gradients are unbiased estimates, particularly during the initial training iterations.

2) *Loss Function*: The loss function will return a higher value if the model makes a bad prediction and a smaller

value if the model makes a good forecast. We have opted for "binary_crossentropy" as the loss function for the suggested structure. The accuracy of a model's outputs is measured using function losses. There needs to be an evaluation of the crossentropy loss among the labels and the forecasts.

3) *Metrics*: Metrics provide a quantitative evaluation of the model's efficacy. The accuracy statistic has been used to evaluate the effectiveness of the suggested model. Table III displays the optimal solution and loss function for the model we have proposed, as well as other functions for consideration in future studies.

TABLE III
PARAMETERS FOR THE PROPOSED MODEL COMPILATION

Parameter	Selected	Alternatives
Optimizer	Adam	RMSprop
Loss Function	binary_crossentropy	Focal Loss

E. Evaluation

Fig. 4 shows a comparison between the training accuracy (green line) and the validation accuracy (red line). We can observe that both of the accuracies have been improved, with the validation accuracy of 0.9828 and the training accuracy of 0.99. In addition, Fig. 5 shows a graph of training and validation loss, where the red line represents increasing validation loss and the green line represents decreasing training loss. The validation loss needs to be lowered which can be accomplished by tweaking the model's layers and enhancing the processing system.

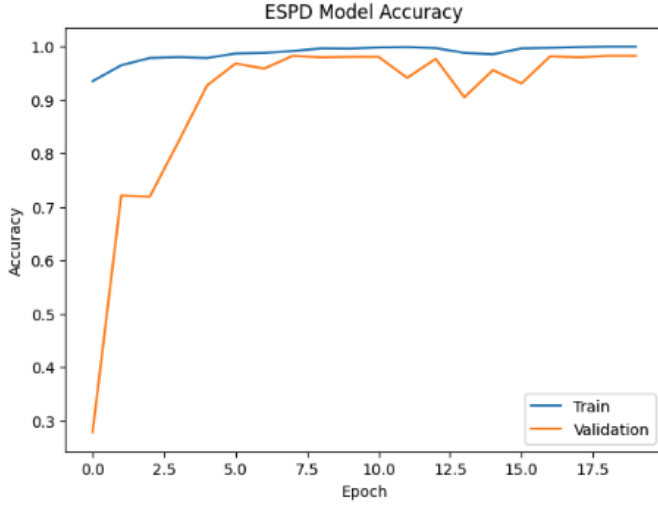


Fig. 4. Evaluation of the Training & Validation Accuracy

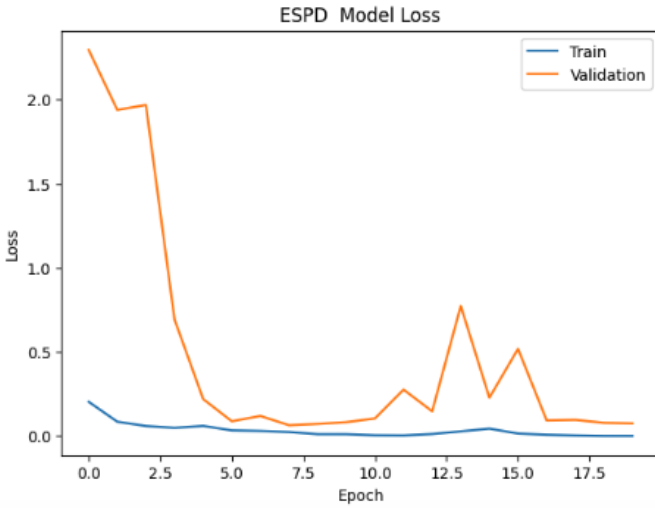


Fig. 5. Evaluation of the Training & Validation Loss

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Evaluation of Each Model for Pneumonia Detection

During evaluating any model, accuracy, data volume and parameters are critical. The accuracy provides an assessment of the model, while the parameter shows the efficacy of the model. We are highlighting the fact that our model falls under the category of lightweight CNNs, despite having a smaller dataset than many of the existing models. Our model surprisingly achieves a high degree of accuracy which illustrates that our method can produce significant performance outcomes in terms of the accuracy. In addition, Table IV shows the standard classification report and Fig. 6 shows the standard confusion matrix of our proposed model.

$$\text{Accuracy (ACC)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

TABLE IV
CLASSIFICATION REPORT OF THE PROPOSED MODEL

Precision	Recall	F1 Score	Specificity	Accuracy
98.22	98.2	98.60	97.01	98.24

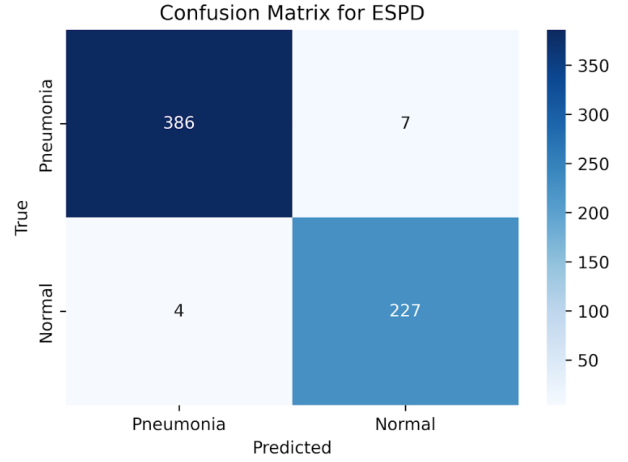


Fig. 6. Confusion Matrix of the Proposed Model

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (3)$$

where Precision is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

,Recall is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

and Specificity is defined as:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (6)$$

The numbers in these equations stand for the number of true positive predictions (TP), true negative predictions (TN), false positive predictions (FP), and false negative predictions (FN). The F1 score offers a fair assessment of precision and recall, while the accuracy shows the percentage of accurate predictions over all forecasts.

B. Model Fitting Analysis

Table V contains the train loss and train accuracy numbers, whereas Table VI contains the validation loss and validation accuracy data. To see the progression, we have collected samples from several epochs. Table V shows a progressive rise in the training accuracy in each epoch, beginning at 0.93 and reaching 0.999. We can also observe from Table VI that shows a progressive rise in the validation accuracy in each epoch, beginning at 0.2787 and reaching 0.9828. Increasing the epoch quantity can enhance the accuracy of a model.

TABLE V
PROGRESSION OF THE MODEL TRAIN

Epoch	Train Loss	Train Accuracy
1	0.2033	0.93
5	0.0599	0.0.978
10	0.0106	0.996
20	9.1025e-05	0.999

TABLE VI
PROGRESSION OF THE MODEL VALIDATION

Epoch	Validation Loss	Validation Accuracy
1	2.2954	0.2787
5	0.2194	0.9272
10	0.0822	0.9808
20	0.0749	0.9828

Table VII depicts that the accuracy of the EPSD model which is utilized in our system seems promising when contrasted with other existing models in the literature. The suggested approach has decreased the validation loss as well as increased the precision of the findings. This demonstrates that the EPSD model is more reliable and solid model. We compared each model to ours in Table VII. Two models are displayed in each column of the table. The first is suggested, and the second is evaluated based on wthe suggested model.

V. CONCLUSION

We have put forth a quick and less expensive computational approach for the automatic diagnosis of Pneumonia. Using the more than 5800 image Pneumonia chest X-ray dataset, a series of studies has been carried out. These investigations aim to categorize the photos into “Pneumonia” and “Normal” situations. By means of these trials, we have demonstrated that, in comparison to the other CNN architectures, the suggested technique offers superior classification performance. If this model is trained on a dataset that has a similar distribution to the data on which the predictions are to be made, it can be used to accurately predict Pneumonia using chest X-rays, so it can be used as a quick and efficient supplementary screening tool for Pneumonia. By further developing this approach, the suggested strategy can potentially address numerous medical issues, including diseases identified through chest X-rays. Real-time application for Pneumonia that is accessible to all is still pending.

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TABLE VII
COMPARISON WITH THE EXISTING MODELS

Existing Works	Model Name	Number of Images	Accuracy
[8]	Proposed Classifier DenseNet201	5,824	97.3% 94.2%
[9]	Superpixel-based histon Fractal dimension	5,856	91.3% 83.3%
[10]	EL VIT	1,00,000	93.91% 92.47%
[11]	Proposed Method DRE-Net	12,531	96.70% 92.59%
[12]	Xception InceptionV3	5,856	96.16% 94.13%
[13]	Proposed DenseNet201	15000	92.12% 69.50%
Proposed	ESPD	5,856	98.24%

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